
Global AI Research Observatory

A Data Warehouse for Analyzing the
International AI Research Boom

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Data Management A.Y. 2025–2026

Research Objective and Analytical Questions

The project integrates scientific publications and socioeconomic indicators into a multidimensional data warehouse for analyzing the international evolution of AI research.

Evolution	How did observed AI research output change from 2018 to 2025?
Geography	Which countries lead AI research, and how does counting method affect the ranking?
Normalization	How do population and economic scale change country comparisons?
Specialization	How do research topics and socioeconomic groups shape the global AI landscape?

Data Sources and Corpus Definition

OpenAlex Scientific publications

- Works and publication dates
- Topics and source taxonomy
- Sources
- Citations and open access

World Bank WDI Socioeconomic context

- Population
- GDP and GDP per capita
- Internet use and R&D expenditure
- 217 economies across 2018–2025



AI CORPUS SELECTION

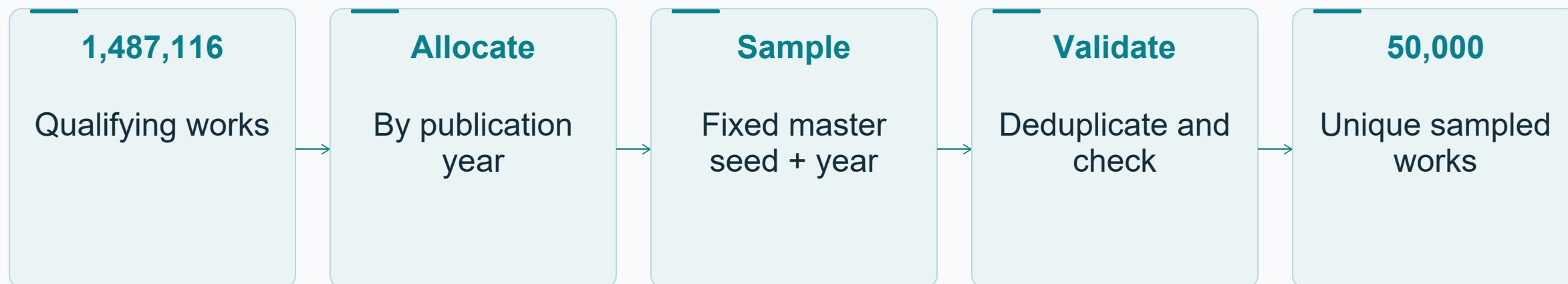
2018-2025 publications → Primary Topic → Artificial Intelligence Subfield (ID 1702)

Corpus membership is determined by the subfield of the Primary Topic, but all associated topics are retained for subsequent thematic analysis.

Six publication types are eligible. Retracted works are excluded.

Reproducible Year-Stratified Sampling

A reproducible year-stratified procedure reduces the qualifying corpus to a tractable analytical sample while preserving coverage across publication years.



Analytical results refer to the sampled corpus, not the complete OpenAlex population.

From Sources to Analytics: End-to-End Architecture

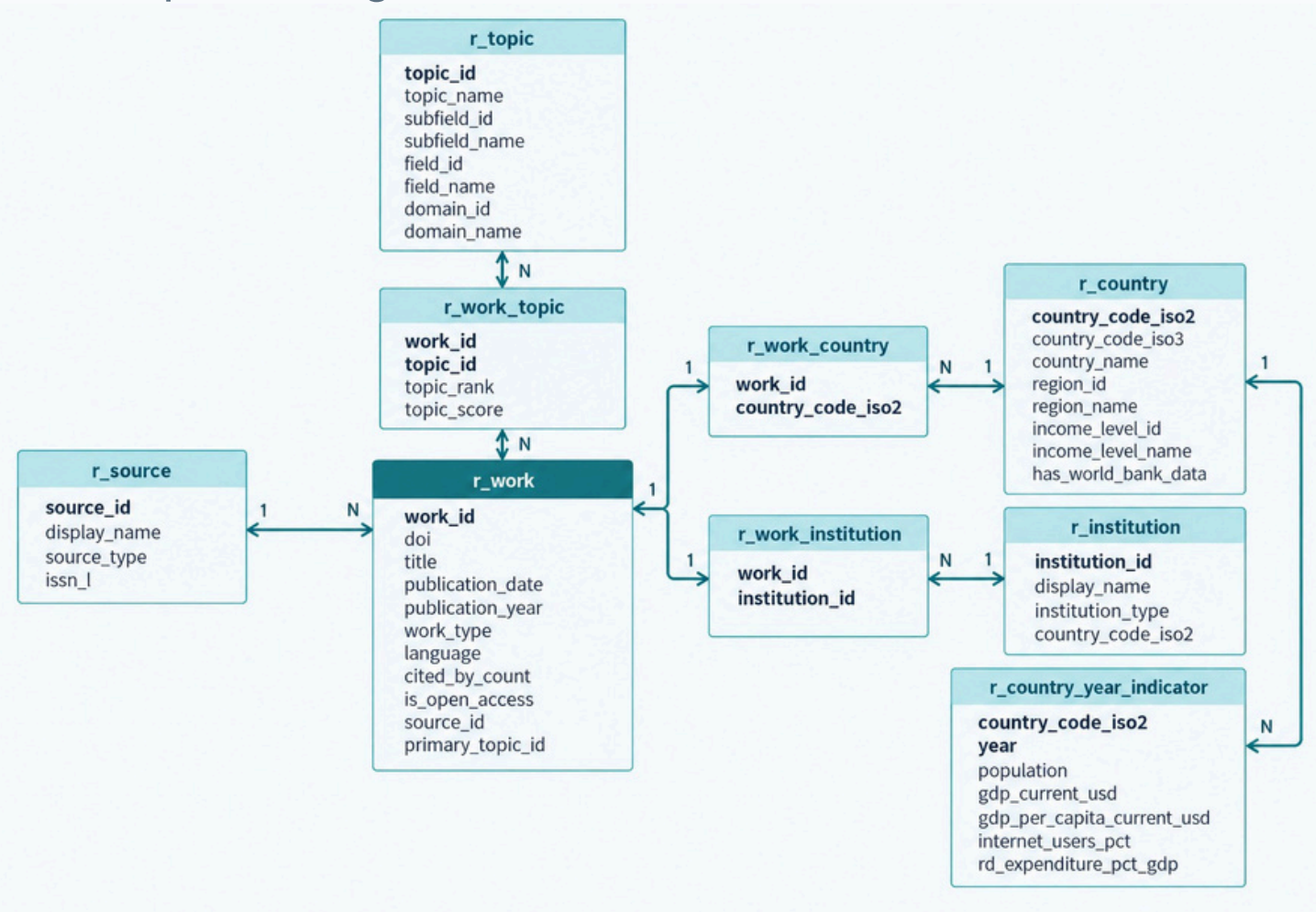
The architecture separates ingestion, reconciliation, dimensional modeling, and analytical consumption into clearly defined layers.



Raw and reconciled data remain separated from the dimensional warehouse used for analytics.

Reconciled Layer: Integrating Heterogeneous Sources

The reconciled layer transforms heterogeneous source data into a consistent, source-preserving relational model.



Preserve source identity

Natural/source keys remain unchanged.

Reconcile relationships

Works, topics, countries, institutions and sources are linked consistently.

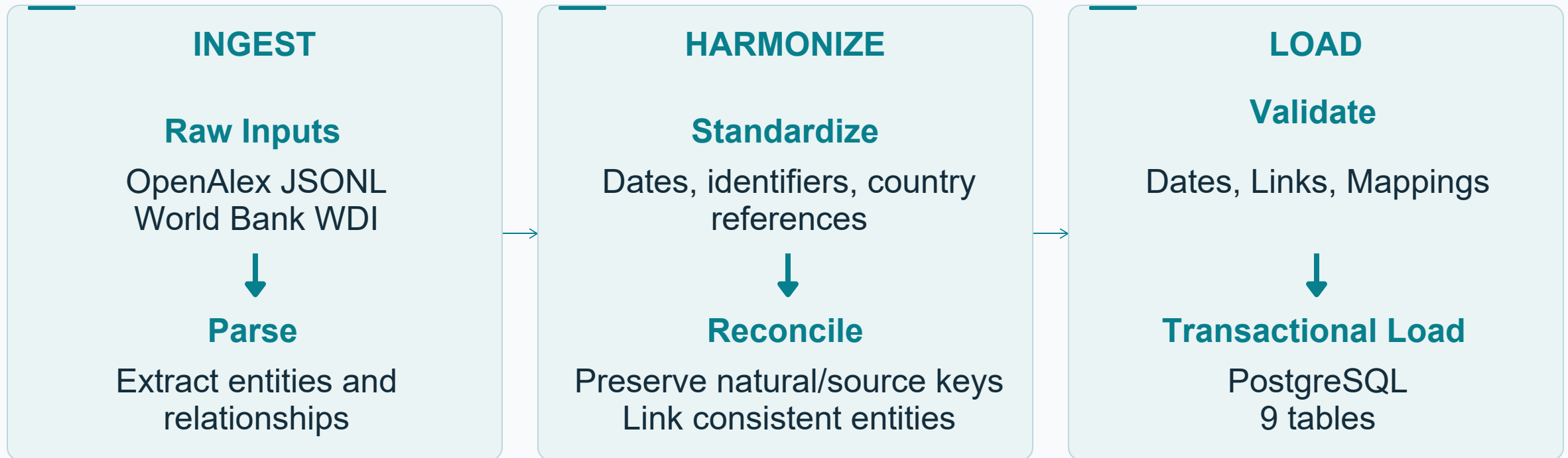
Validate before loading

Invalid dates, mappings and relationships are rejected before warehouse loading.

Selected attributes shown; technical fields are omitted for readability.

ETL and Reconciliation Pipeline

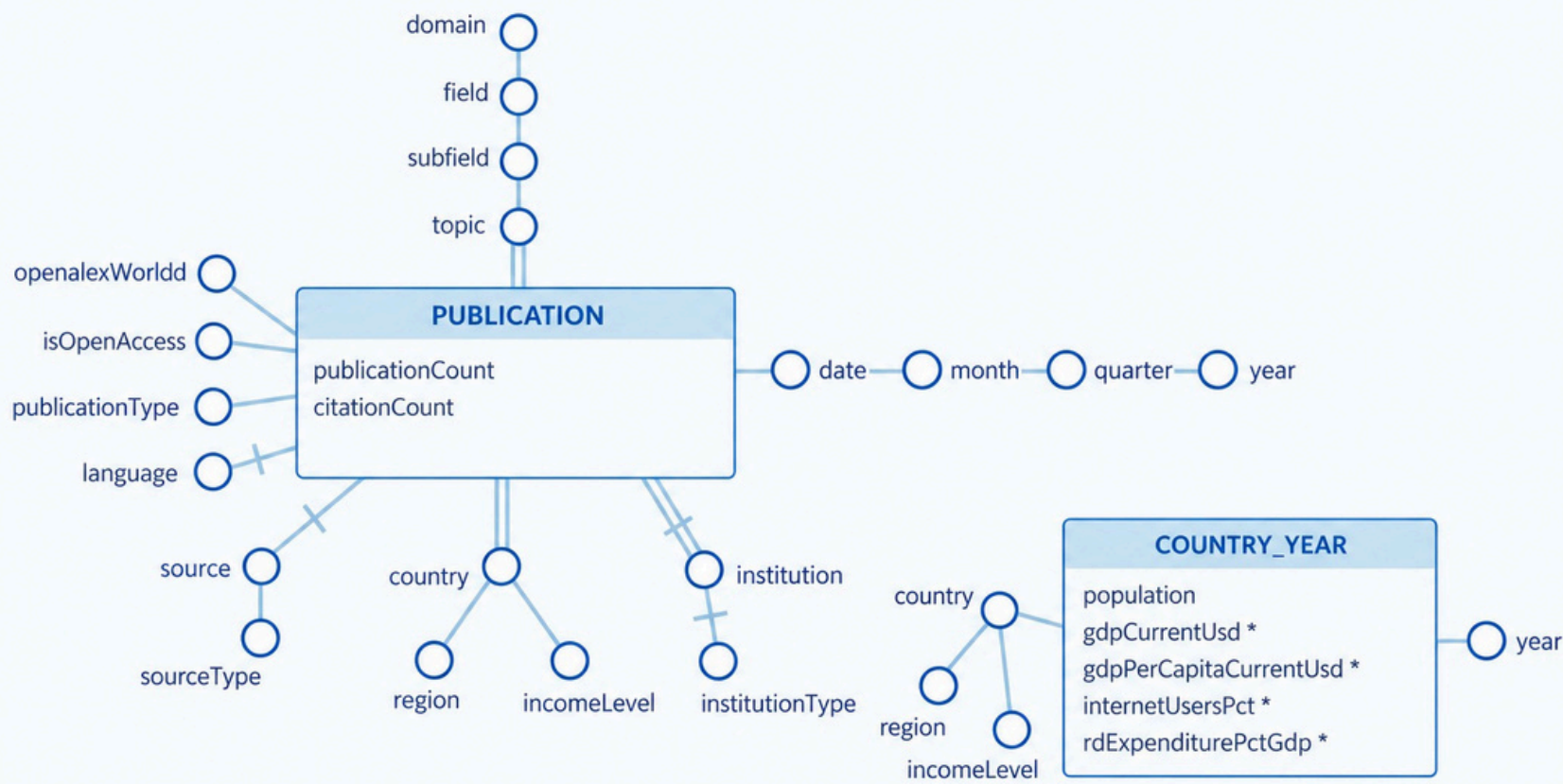
Raw OpenAlex and World Bank data are parsed, standardized, reconciled, validated, and transactionally loaded into PostgreSQL.



Missing optional relationships → no artificial members · Missing WDI values → SQL NULL

Conceptual Multidimensional Model - DFM

The conceptual model identifies two analytical facts with different grains and their associated dimensions and hierarchies.



Two Analytical Facts

Publication
Country × Year

Explicit Grains

One sampled publication
One observed economy × year

Hierarchical Dimensions

Time · Topic · Geography
Source · Institution

Selected attributes shown; technical fields are omitted for readability.

Measures, Additivity and Missing Data

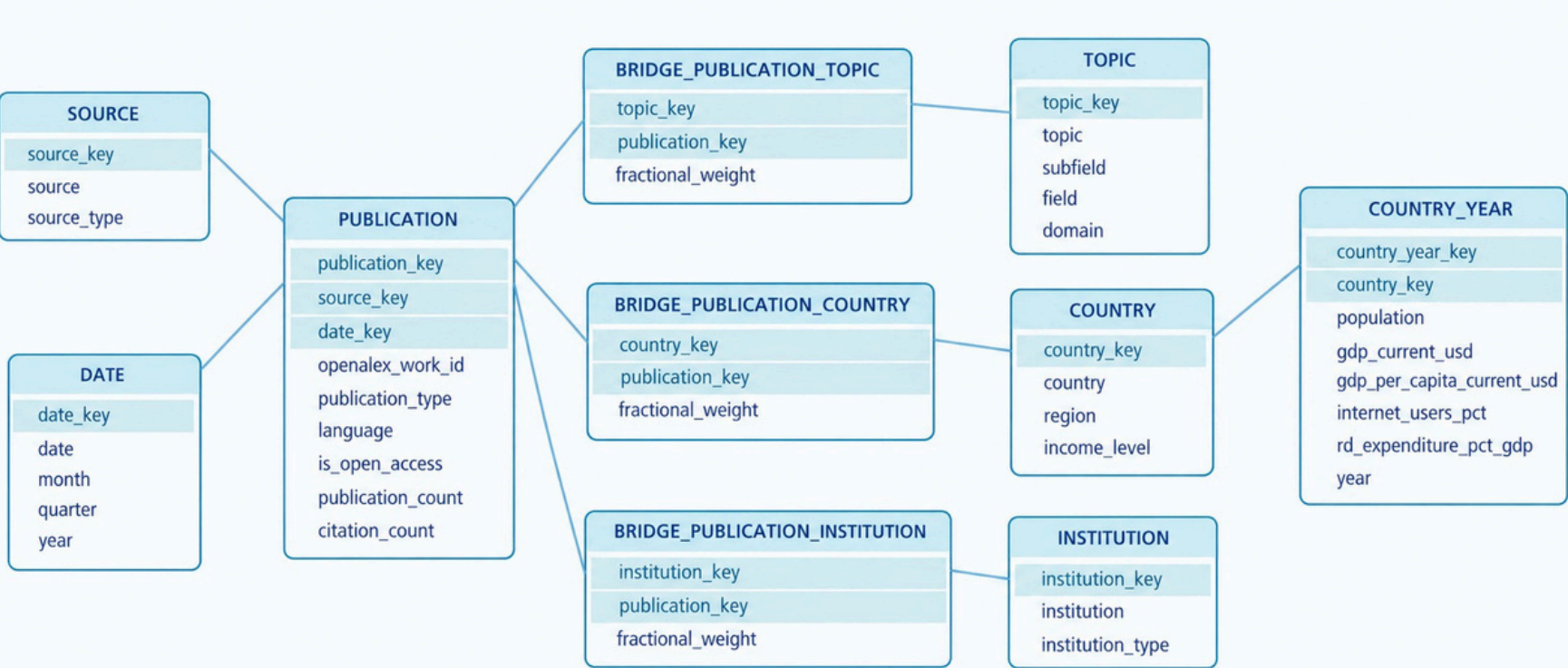
Each measure has specific aggregation semantics that must be preserved during OLAP analysis.

Measure	Semantics · Aggregation
Publication count	Event count · Additive at publication grain
Citation count	Cumulative · Additive across publications; Age-biased over time
Population	Level measure · Non-additive over time
GDP	Annual flow measure · Not directly additive across years
GDP per capita	Unit measure · Non-additive
Internet users (%)	Percentage measure · Non-additive
R&D expenditure (% GDP)	Percentage measure · Non-additive

Missing WDI values remain NULL. No imputation. Missing or zero denominators yield NULL.

Physical Warehouse: Fact Constellation

The conceptual DFM is implemented as a PostgreSQL fact constellation with two facts, shared dimensions, and bridge tables for multivalued relationships.



Two Analytical Facts

Publication
Country × Year

Three Bridge Tables

Topic · Country · Institution
Fractional weights

Conformed Geography

dim_country shared
across facts
Separate time grains

Selected attributes shown; technical fields are omitted for readability.

Bridge-Aware Counting and Cross-Fact Analysis

Full and Fractional Counting

One publication linked to USA and UK

FULL
USA = 1
UK = 1

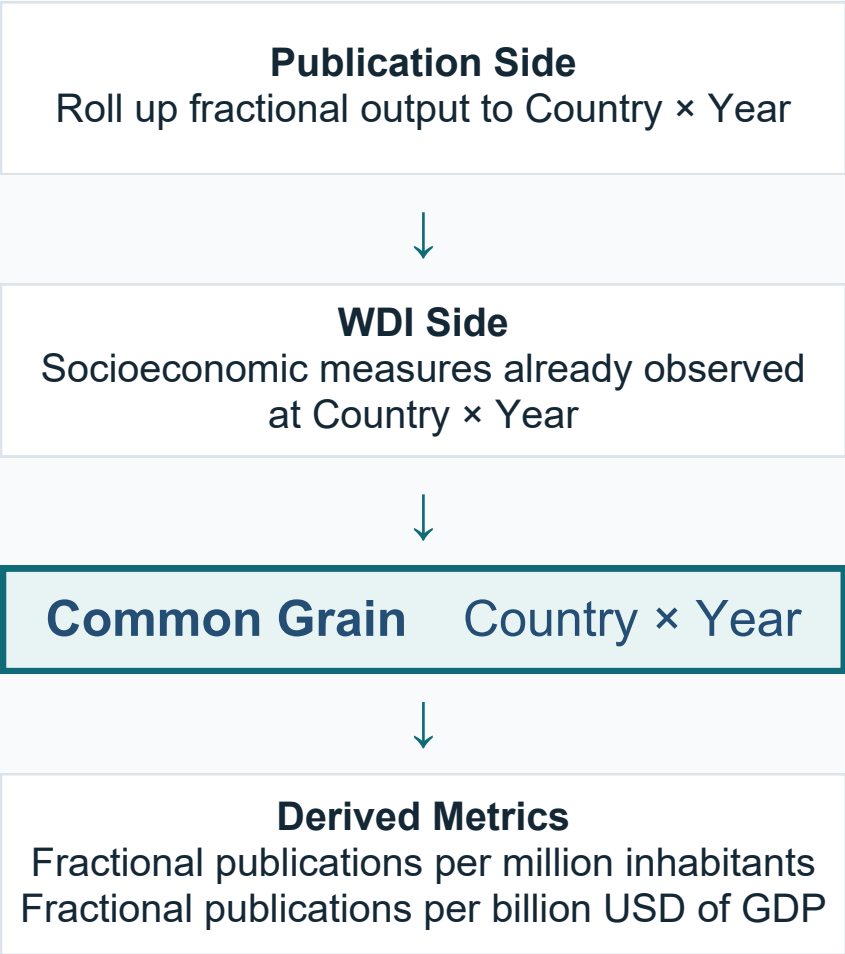
Full credit

FRACTIONAL
USA = 0.5
UK = 0.5

Proportional attribution

Fractional weights sum to 1 for each publication within a bridge.

Drill-Across at Country × Year



Missing or zero denominator → NULL

OLAP Session Design

OLAP Sessions

01 - Research Growth 2018–2025

Annual evolution and adjacent-period comparison

02 - Geographic Full vs Fractional

Country rankings under alternative attribution semantics

03 - Normalized Leadership

Fractional publications per million · full support ≥ 30

04 - Income-Level Fractional Share

Country-attributed output by income group over time

05 - Topic Annual Shares

Thematic composition across years

06 - Wealth vs AI Intensity

GDP per capita vs fractional publications per million

OLAP Capabilities

Temporal analysis

Year slice · roll-up · adjacent-period comparison

Ranking & attribution

Full vs Fractional counting

Normalization

Country $\times$ Year drill-across · per-capita metrics

Hierarchical aggregation

Country $\rightarrow$ Income Level

Thematic analysis

Topic slicing across years

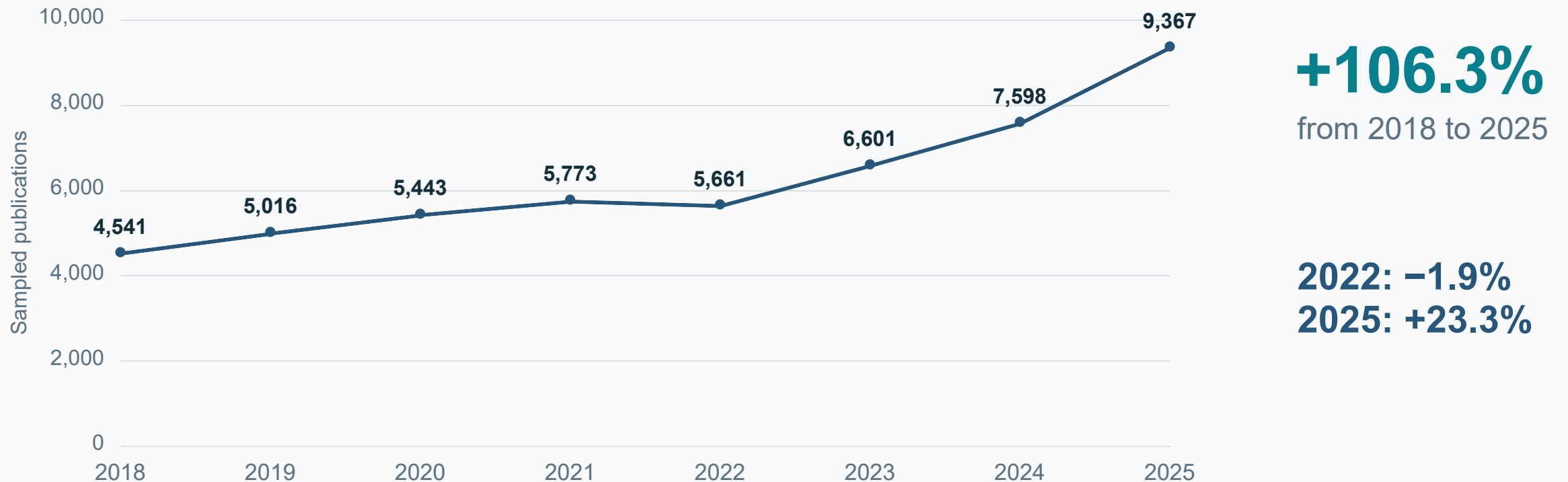
Cross-fact comparison

Publication intensity $\leftrightarrow$ socioeconomic measures

Each session corresponds to one or more OLAP operations over the warehouse schema.

AI Research Output More Than Doubled Between 2018 and 2025

Analyze the yearly evolution of sampled AI publications across the 2018–2025 time dimension.



Growth was uneven: output dipped slightly in 2022, then accelerated sharply in 2023–2025.

Attribution Semantics Change Country Rankings Below the Top

Assess how country rankings change under full versus fractional publication attribution.

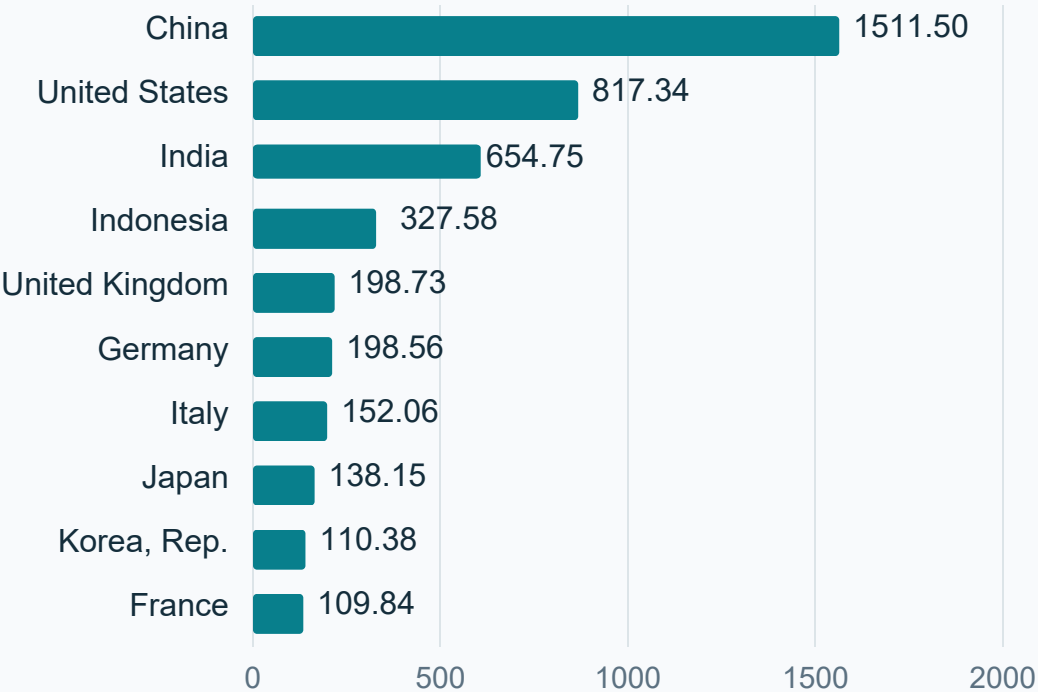
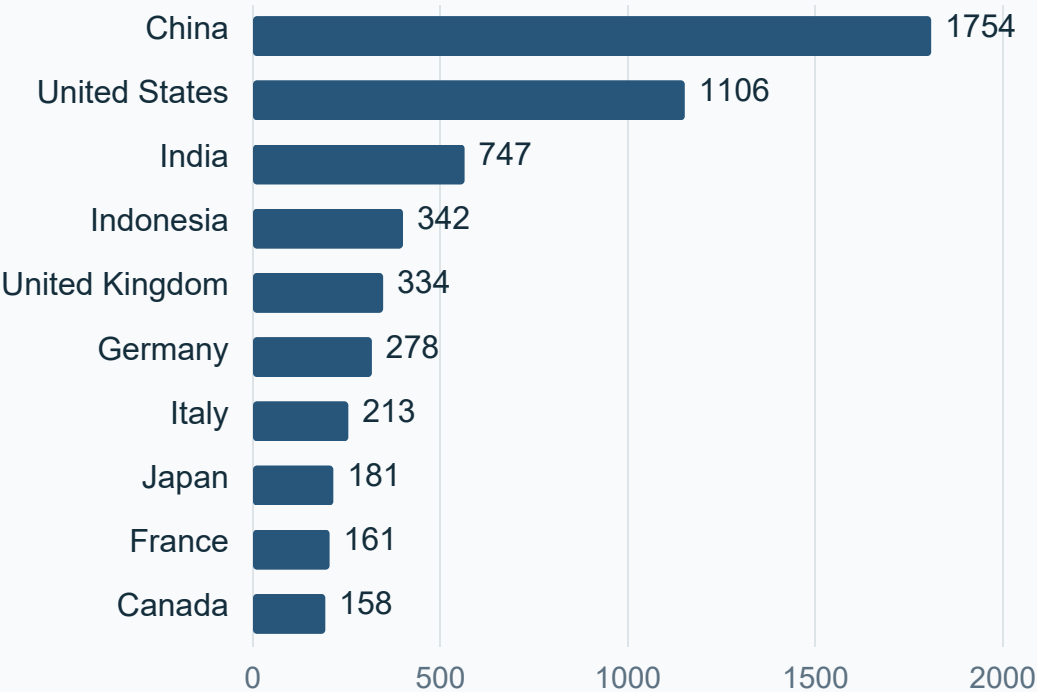
Full participation

Each country receives full credit for every publication in which it participates.

Fractional attribution

Each publication is proportionally distributed across its unique associated countries.

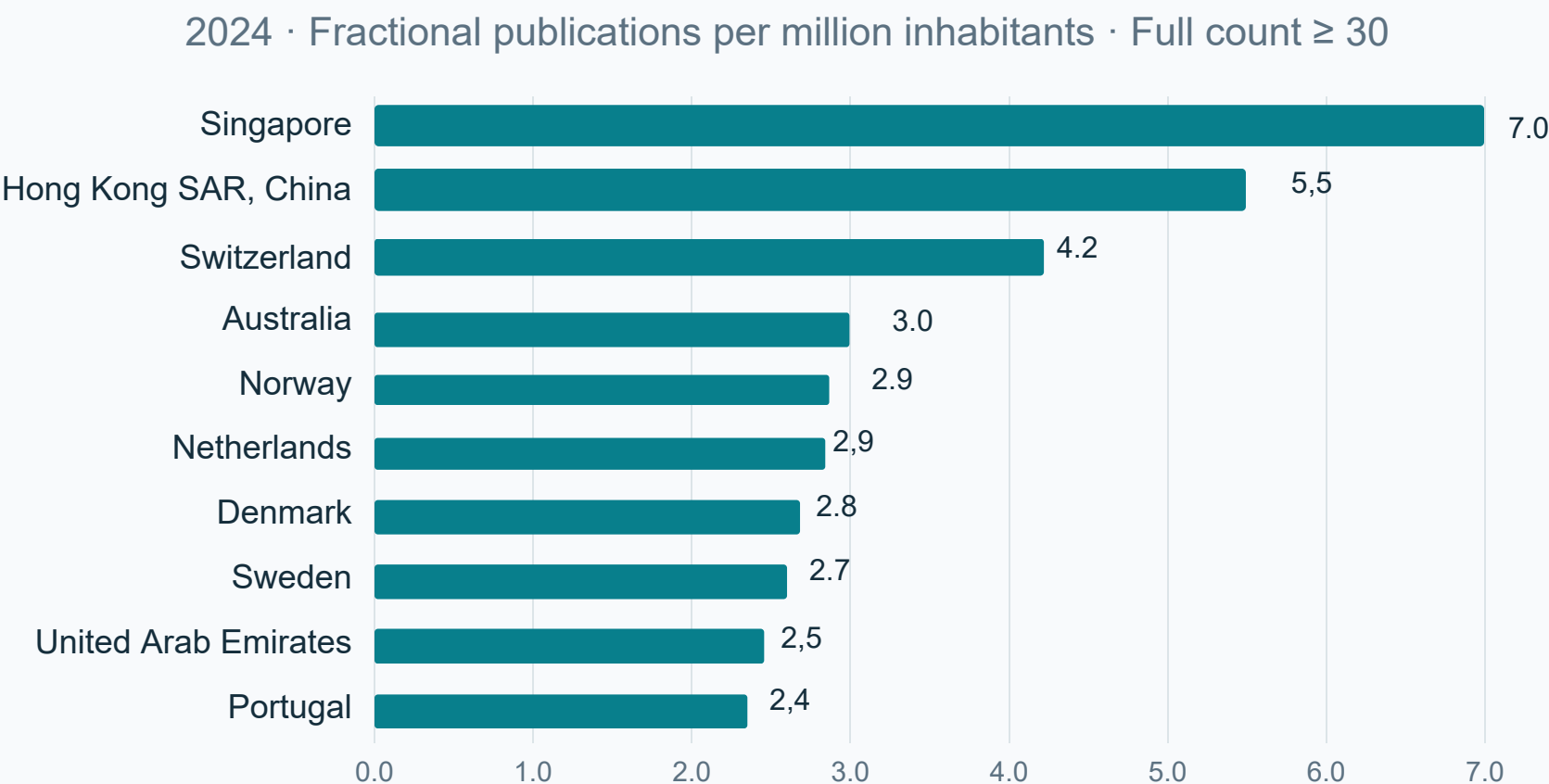
Country ranking · 2025 slice



Top positions remain stable across counting methods, while fractional attribution reshuffles the lower ranks.

Normalizing for National Scale Changes the Leadership Picture

Assess how country leadership changes after normalizing fractional output by population.



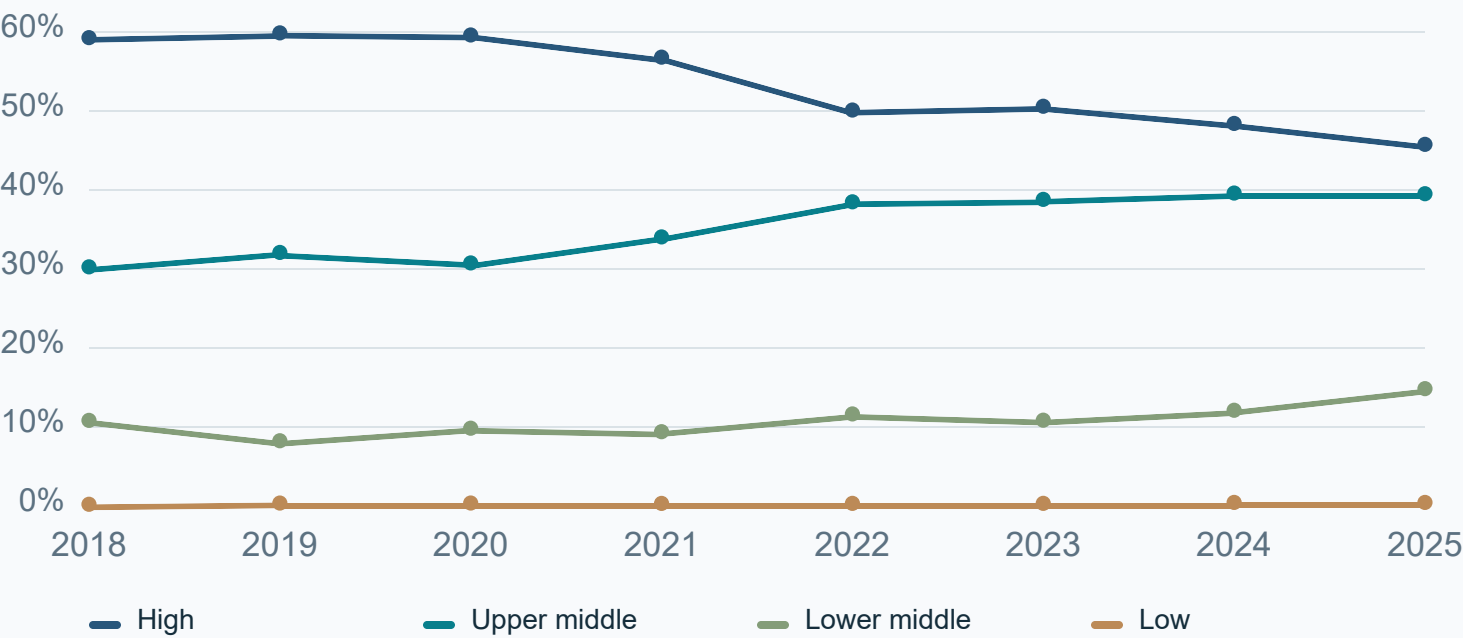
SUPPORT FILTER
Full count ≥ 30
Descriptive stability
threshold · not a
significance test

High absolute output does not necessarily imply high per-capita research intensity.

Middle-Income Economies Gain Share, but the Research-Intensity Gap Persists

Assess how the distribution and research intensity of AI output differ across World Bank income groups.

Annual share of known-income output



Income groups use the extraction-time country classification consistently across 2018–2025

2024 publications per million

Income group	Intensity
High	1.702
Upper middle	0.632
Lower middle	0.203
Low	0.021

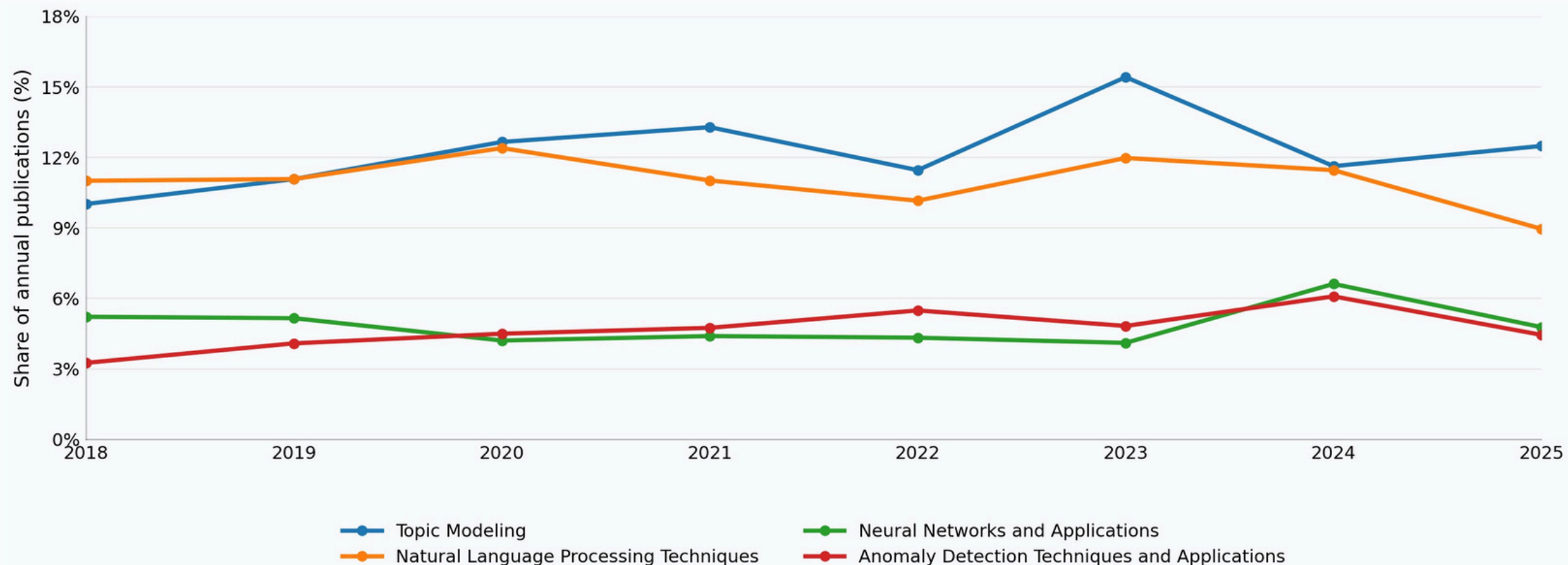
Matched population coverage. Low-income intensity is positive.

Middle-income groups gain share of attributed output, but high-income economies retain a much higher research intensity.

The Thematic Composition of AI Research Changes Over Time

Assess how the relative prominence of Top 4 topics by cumulative publication participation changes over time.

2018–2025 · Full topic participation · Non-exclusive topic shares



The leading topics remain prominent throughout the period, but their relative annual shares evolve differently.

AI Research Intensity Tends to Increase with GDP per Capita

Assess how AI research intensity varies with national economic prosperity.

2024 country slice · Fractional publications per million · Full count ≥ 30 · 37 complete country pairs



37 COUNTRIES

Complete 2024 pairs
X GDP per capita
Y Fractional
publications / million
Log-scaled X axis

Countries with higher GDP per capita generally show greater AI research intensity, although substantial variation remains.

Key Findings, Limitations and Live Demo

The warehouse enables reproducible multidimensional analysis while keeping methodological limitations explicit.

Key findings

- Observed AI research output more than doubles from 2018 to 2025
- Metric choice changes country rankings: fractional counting reshuffles positions, while population normalization changes research intensity.
- Middle-income participation rises while research-intensity gaps persist
- Leading topic shares follow different trajectories over time
- AI research intensity is positively associated with GDP per capita.

Limitations

- 50,000-work sample, not the full OpenAlex corpus
- OpenAlex metadata evolves over time
- WDI coverage varies across indicators
- Normalized metrics depend on denominator choice
- Topic memberships are non-exclusive
- Geographic analyses are limited to works with identifiable country attribution (64.8%); unattributed works are excluded.

NEXT
5-minute live demo